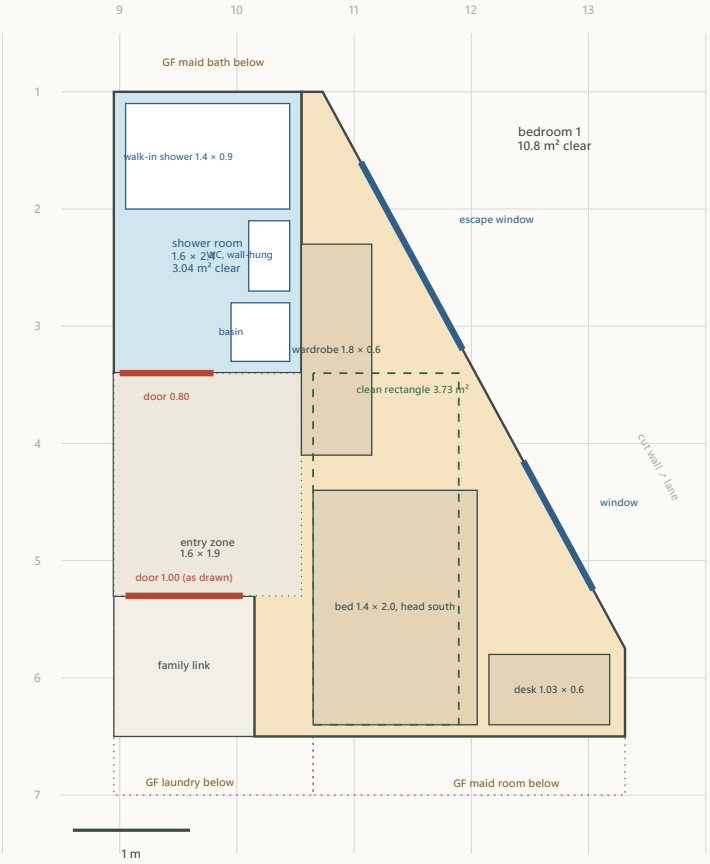



First floor, CSV frame. Blue: the shower room and its fixtures; stone: the furniture; red: the doors; blue bars: the windows on the cut wall; dotted: the ground-floor rooms below; green dashed: the bedroom's clean rectangle (east of the entry zone, south of the cut's narrow part). The cut wall at its 6 Nov 2026 position (0.12 m further out).

# Bedroom 1 with a simple en-suite shower room · short study

NOT ADOPTED: young children's room (owner, V9.0)

**The brief:** a shower, a WC and a basin, wall-hung on one service wall, over the laundry / maid wing below, in the east cluster; the bedroom keeps its bed 1.4 × 2.0, a wardrobe ≥ 1.8 m, a desk, its door and its escape window. **The best layout found (of the positions tried in `studies/bedroom1_bath.typ`, now with the cut wall 0.12 m further out):** the shower room in the room's north-west corner (1.6 × 2.4 over the maid bath and the laundry), its service wall on the bedroom side; the room's existing door (y 5.3) then opens into an entry zone of 1.6 × 1.9 m between the shower room and the bed; the bedroom proper lies east of it, under the cut wall.

| Item | m² | Note |
|---|---|---|
| Bedroom 1 as drawn today (no bath), clear | 14.64 | on axes 16.42 the cut wall's move added 0.79 |
| Shower room 1.6 × 2.4, clear | 3.04 | walk-in shower 1.4 × 0.9 at the north end, WC and basin on the service wall x 10.55; the stack drops into the laundry / maid bath, east cluster; mechanical extract (no window) |
| **Bedroom proper, clear** (the room minus the shower room, the entry zone included) | **10.80** | ≥ 10 m²: the brief's area is met |
| … of which the clean rectangle (x 10.65–11.89, y 3.4–6.4) | 3.73 | 1.24 × 3.00 m: the rest is the wedge under the cut wall and the entry zone |
| Bed 1.4 × 2.0, head on the south wall (x 10.65–12.05, y 4.4–6.4) | 2.80 | against the shower-room wall on its west side; on its east side 0.39 m to the cut's inner face at the head end, widening to 1.14 m at the foot; 1.00 m between the foot and the wardrobe line |
| Wardrobe 1.8 × 0.6 (x 10.55–11.15, y 2.3–4.1) | 1.08 | in the wedge, backing onto the shower-room wall; its north end 0.14 m inside the cut's inner face |
| Desk 1.03 × 0.6 (x 12.15–13.18, y 5.8–6.4) | 0.62 | on the south wall east of the bed, under the second window's light; shorter than the usual 1.2 m |
| Doors · window | — | the room's door 1.00 m at y 5.3 unchanged; the shower-room door 0.80 m in its south wall, off the entry zone; the escape window (1.82 m on the cut, sill 0.9) stays, now facing the wardrobe's end and the wedge; the second window lights the bed |

**Study checks passed** (5)

## Does it work? Plainly:

**By the numbers, yes:** the shower room fits over wet rooms with its stack in the east cluster, and the bedroom keeps 10.80 m² clear with the bed, a 1.8 m wardrobe and a desk, the door and the escape window unchanged. **As a room, barely:** the clean rectangle left for living in is only 1.24 × 3.00 m (3.73 m²), the bed stands against the shower-room wall with 0.39 m beside its head on the other side, the wardrobe and the entry zone sit in the wedge, and the desk shrinks to 1.03 m. It is a hotel-style small room, not the clean 10 m² room the earlier brief asked for; the cut wall's extra 0.12 m helped (+0.79 m²) but did not change the shape.
**Recommendation:** keep option A of 2 Nov 2026 (no private bath: the family bath is 3.5 m from the door, night lights on the route) unless the owner values the en-suite more than the room's shape; if the en-suite is wanted, this is the layout to build, and a further cut-wall move (0.14 m) does not change the picture. Not applied.

Bedroom 1 en-suite study · the room from data/rooms_ff.csv (6 Nov 2026), the link and the door from studies/ff_family.typ, the ground-floor rooms from data/rooms_gf.csv; the brief and the measures from studies/bedroom1_bath.typ (clear ≈ area − 0.1 × perimeter). Checks: the shower room lies over wet / service rooms only (the maid bath y 1–2.8, the laundry 2.8–3.40 west of the maid room's partition x 10.65): its stack joins the east cluster · the shower room 1.6 × 2.4 holds the walk-in shower, the wall-hung WC and the basin on one service wall (x 10.55) without overlap · the bedroom proper keeps ≥ 10 m² clear (10.80 m²) with the bed 1.4 × 2.0, a 1.8 m wardrobe and a desk placed without overlap and inside the cut's inner face · the room's door (y 5.3, 1.00 m) and the escape window on the cut stay as drawn; the shower-room door 0.80 opens off the entry zone, not into the bed's space · the clean rectangle of the bedroom proper (1.24 × 3.00 = 3.73 m²) is SMALLER than the 10 m² the owner's brief asked for: the room works on paper, not as a clean room.